

A Project Report on

AUTOMATED PODCAST TRANSCRIPTION AND TOPIC SEGMENTATION

This project is submitted as a requirement for the successful completion of
the
virtual internship program, delivered through the *Infosys springboard*
Virtual Internship program

In
ARTIFICIAL INTELLIGENCE

Submitted by
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--Under the Guidance of--
Goutham Nishkal
Project Mentor

CERTIFICATE

This is to certify that the project titled “**AUTOMATED PODCAST TRANSCRIPTION AND TOPIC SEGMENTATION**” is a bonafide work carried out by NIDHI PASWAN as a part of the Infosys Springboard Virtual Internship Program in the domain of ARTIFICIAL INTELLIGENCE.

This project has been completed under my guidance and supervision and fulfills the requirements for the successful completion of the virtual internship program delivered through Infosys Springboard.

To the best of my knowledge, the work presented in this project report is original and has not been copied or plagiarized from any source. The project demonstrates the candidate’s understanding, skills, and practical application of concepts related to Artificial Intelligence, Machine Learning, and Natural Language Processing.

I wish her all the best for her future academic and professional endeavors.

Under the Guidance of

Mr. Goutham Nishkal

Project Mentor

Infosys Springboard Virtual Internship Program

CANDIDATE'S DECLARATION

I hereby declare that the project entitled “**Automated Podcast Transcription and Topic Segmentation using Artificial Intelligence**” is an original work carried out by me under the **Infosys Springboard Virtual Internship Program**. This project has been completed by me under the guidance and learning resources provided by Infosys Springboard.

I confirm that this project work is **entirely my own effort** and has **not been copied or plagiarized** from any source. All the information, data, code, and references used in this project are either self-developed or properly acknowledged wherever applicable.

I further declare that this project has **not been submitted previously** to any other university, institution, or organization for the award of any degree, diploma, or certification.

I take full responsibility for the originality, authenticity, and correctness of the content presented in this project report.

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Internship Platform: Infosys Springboard

Project Title: Automated Podcast Transcription and Topic Segmentation

Duration: Virtual Internship

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Infosys Springboard** for providing me the opportunity to undertake this virtual internship and work on the project titled “**Automated Podcast Transcription and Topic Segmentation.**” This internship has helped me gain practical exposure to real-world applications of Artificial Intelligence and Machine Learning.

I am extremely thankful to my project mentor **Mr. Goutham Nishkal Deepu** for his valuable guidance, constant encouragement, and insightful feedback throughout the duration of this project. His support played a crucial role in the successful completion of this work.

I would also like to acknowledge the efforts of the **Infosys Springboard Virtual Internship Program Team** for designing such a well-structured learning platform and providing necessary resources that enhanced my technical and professional skills.

I extend my heartfelt thanks to my teachers, peers, and all those who directly or indirectly supported me during the development of this project. Their motivation and cooperation helped me overcome challenges and complete the project successfully.

Finally, I am grateful to my family for their continuous support, encouragement, and motivation throughout this learning journey.

TABLE OF CONTENTS

1. Certificate.....	2
2. Candidate declaration.....	3
3. Acknowledgement.....	4
4. Abstraction.....	6
5.Introduction.....	7
6. Problem Statement	8
7. Problem and Solution.....	9
8. Methodology.....	10-13
9. Flow Diagram.....	14-15
10. Technologies Used.....	16
11. Implementation.....	17-19
12. Use Cases.....	20-21
13. Advantages.....	22
14 Limitations.....	22
15. Future Enhancement.....	22
16. Conclusion.....	23

ABSTRACT

With the rapid growth of digital media, podcasts have become an important source of information and knowledge sharing across various domains such as healthcare, education, and technology. However, manually transcribing and analyzing long podcast episodes is a time-consuming and inefficient process. This project, “**Automated Podcast Transcription and Topic Segmentation,**” aims to develop an intelligent system that automatically converts podcast audio into text and segments the content into meaningful topics.

The proposed system accepts podcast audio as input, preprocesses it to improve audio quality, transcribes the speech into text using speech-to-text techniques, and further applies Natural Language Processing (NLP) methods to segment the transcript into coherent topic-based sections. Additionally, the system extracts keywords, generates summaries, and provides an interactive user interface for better readability and understanding.

This project reduces manual effort, improves accessibility of audio content, and enables efficient information retrieval from long podcast episodes. The system is implemented using Python, Flask, machine learning models, and a React-based frontend, making it suitable for real-world applications in media analysis, education, and content management systems.

Keywords

Speech-to-Text, Podcast Analysis, Natural Language Processing, Deep Learning, Topic Segmentation, Keyword Extraction, Text Summarization, Audio Processing

INTRODUCTION

In recent years, podcasts have emerged as a popular medium for sharing information, opinions, and expert discussions across various fields. Millions of podcast episodes are published daily, covering topics such as technology, healthcare, education, and entertainment. Despite their popularity, podcasts remain largely unstructured audio data, making it difficult for users to quickly locate relevant information.

Manual transcription and analysis of podcast content is not only time-consuming but also prone to human errors. There is a growing need for automated systems that can convert speech into text and organize the content into meaningful segments for easy navigation and understanding.

This project focuses on building an **automated podcast transcription and topic segmentation system** that leverages advances in speech technology and natural language processing. The system automatically transcribes podcast audio, identifies topic boundaries, extracts important keywords, and generates summaries. A user-friendly interface is also provided to visualize the results effectively.

The proposed solution enhances accessibility for hearing-impaired users, improves content searchability, and saves significant manual effort, making it a valuable tool for researchers, students, and content creators.

PROBLEM STATEMENT

Podcasts contain valuable and insightful information; however, they are primarily available in audio format, which poses several challenges:

- Difficulty in searching specific information within long audio files
- Lack of structured content representation.
- Time-consuming manual transcription and analysis.
- Limited accessibility for hearing-impaired users.
- Inefficient content reuse and summarization.

Existing solutions either focus only on transcription or provide limited segmentation capabilities, making them insufficient for comprehensive content analysis.

Therefore, the problem is to design and develop an automated system that can accurately transcribe podcast audio and segment the transcript into meaningful topics while providing keyword extraction and summary generation to enhance content usability and accessibility.

OBJECTIVE

The main objectives of the project “**Automated Podcast Transcription and Topic Segmentation**” are:

- To develop an automated system for **transcribing podcast audio into accurate text** using speech recognition techniques.
- To apply **audio preprocessing and noise reduction** methods to improve transcription quality
- To perform **topic segmentation and keyword extraction** from the transcribed text using Natural Language Processing.
- To generate **concise summaries** for each podcast segment to improve content understanding.
- To design a **user-friendly web interface** for uploading audio, playing podcasts, and viewing segmented results with timestamps.

METHODOLOGY

The proposed methodology focuses on building an **AI-based system for automated podcast transcription, topic segmentation, keyword extraction, and summarization** using **Speech Recognition, Natural Language Processing (NLP), and Deep Learning techniques**. The System is designed using a modular and step by step approach to insure the accuracy, efficiency, and ease of use. The implementation is carried out using a real world podcast and audio obtained from **The American Life Podcast Dataset**, which provide conversational and long-form audio content suitable for speech and language analysis

The complete working of the system is divided into the following stages:

1. Podcast Audio Collection

- The system collects podcast audio samples from **The American Life Podcast Dataset**, which features real-world storytelling, interviews, and conversations.
- This dataset reflects natural speech patterns, making it ideal for speech recognition and NLP analysis.
- Users upload pre-recorded podcast episodes through a web-based interface.
- The system supports commonly used audio formats such as **MP3, WAV, and M4A** for user convenience.
- The collected audio data serves as the primary input for transcription, segmentation, and summarization processes.

Dataset Used:

- *The American Life Podcast Dataset* (publicly available podcast audio samples)
- User-uploaded podcast recordings for testing and evaluation

2. Audio Preprocessing and Noise Reduction

Uploaded audio files often contain background noise and inconsistent audio levels.

To address this, the system performs:

- Audio format conversion (MP3 → WAV)
- Noise reduction using signal processing techniques
- Standardization to **16 kHz mono audio**

This step ensures better accuracy in transcription.

3. Speech-to-Text Transcription

After preprocessing, the cleaned audio is passed to an **Automatic Speech Recognition (ASR)** model.

- A **deep learning-based speech recognition model** is used to convert speech into text
- The output is a **complete transcript** of the podcast
- Time-aligned transcription helps in later segmentation

The generated transcript is stored as a text file for further processing.

4. Text Processing using NLP

The transcribed text is processed using **Natural Language Processing (NLP)** techniques such as:

- Sentence tokenization
- Stop-word removal
- Text normalization

This prepares the transcript for meaningful topic segmentation and analysis.

5. Topic Segmentation

The system divides the podcast transcript into **multiple meaningful segments** based on topic changes.

Each segment contains:

- Start time and end time
- Segment-wise text
- Associated keywords

This helps users understand **different discussion sections** within a single podcast episode.

6. Keyword Extraction and Summarization

For each segmented portion:

- Important **keywords** are extracted using NLP techniques
- A **concise summary** is generated for each segment

This allows users to quickly grasp the main ideas without listening to the entire podcast.

7. Data Storage and Result Generation

The processed outputs are stored in structured formats such as:

- JSON files for segments
- Text files for transcripts

These results are sent to the frontend for visualization.

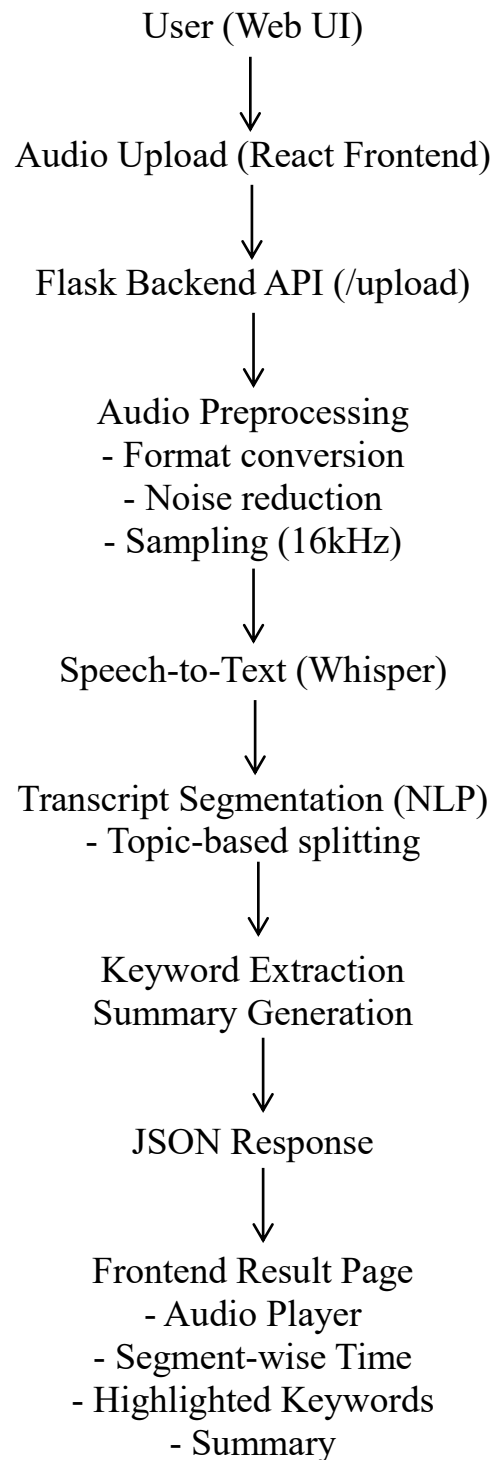
8. User Interface and Output Display

Finally, the system displays:

- Audio player for podcast playback
- Topic-wise segments with timestamps
- Highlighted keywords
- Summaries for each segment

The user-friendly interface allows seamless interaction and easy understanding of podcast content.

FLOW DIAGRAM



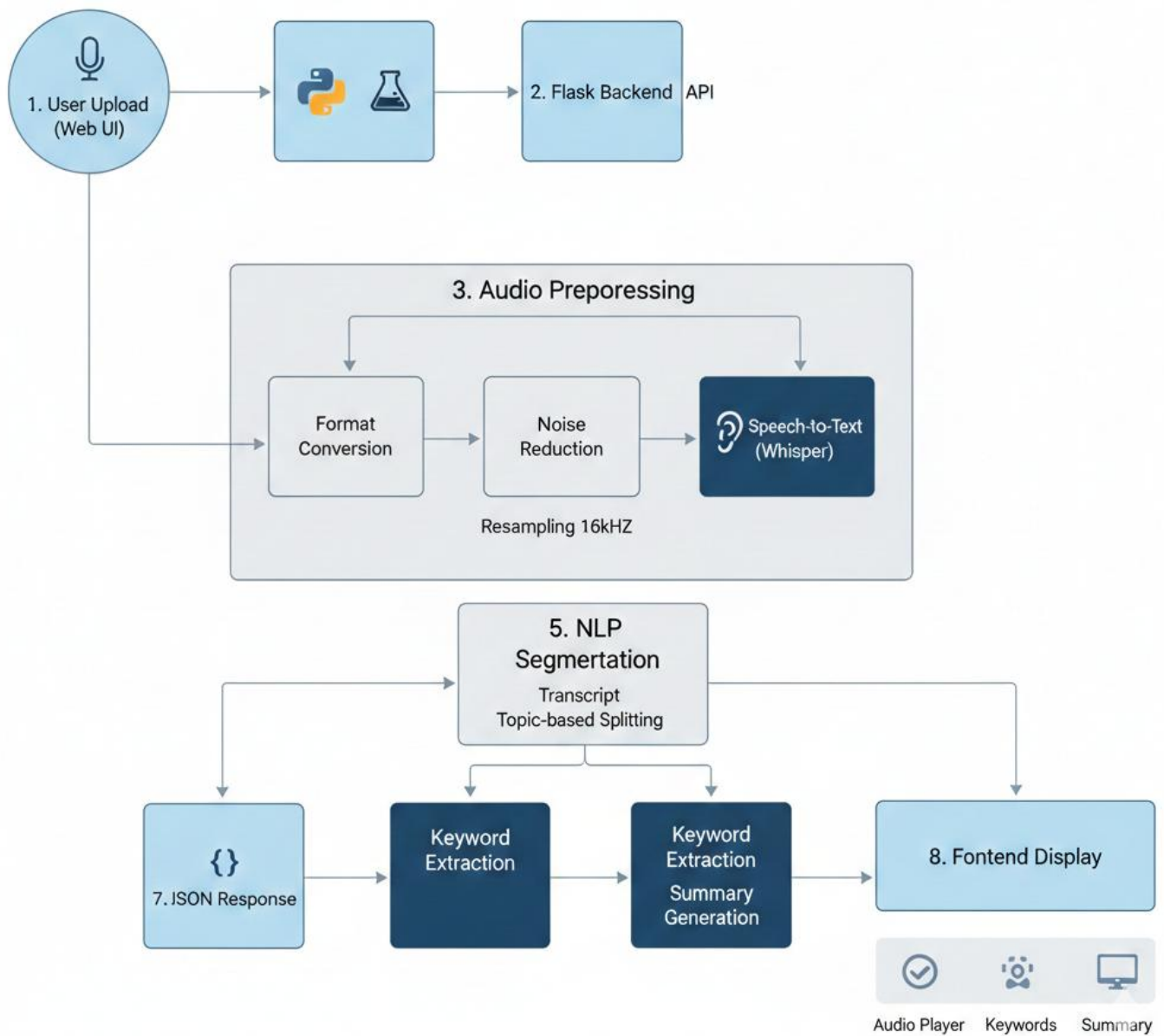
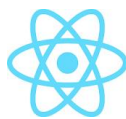


Fig. 1 System Architecture

TECHNOLOGIES USED

The software stack is categorized to show the separation between frontend, backend, and the AI/NLP engine.

Category	Technology	Purpose /Use Case
Frontend	React, Tailwind CSS	Building a responsive UI and managing real-time state.
Backend	Flask	Orchestrating the API, handling file uploads, and routing.
AI Models	Whisper, BART	Speech-to-Text (Whisper) and Abstractive Summarization (BART).
NLP	NLTK, Sentence Transformers	Tokenization, keyword extraction, and topic clustering.
Audio	FFmpeg, Librosa	Audio format conversion, resampling (16kHz), and noise cleaning.
Tools	VSCode, Git	Development environment and version control.

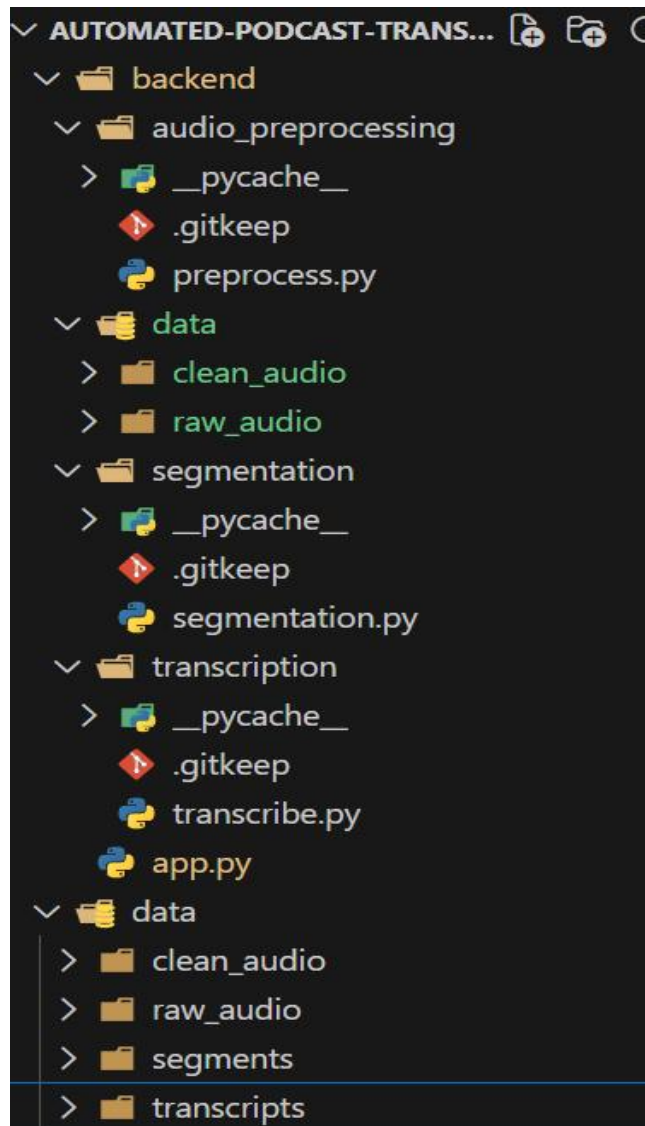


IMPLEMENTATION DETAILS

Backend Modules (Python/Flask):

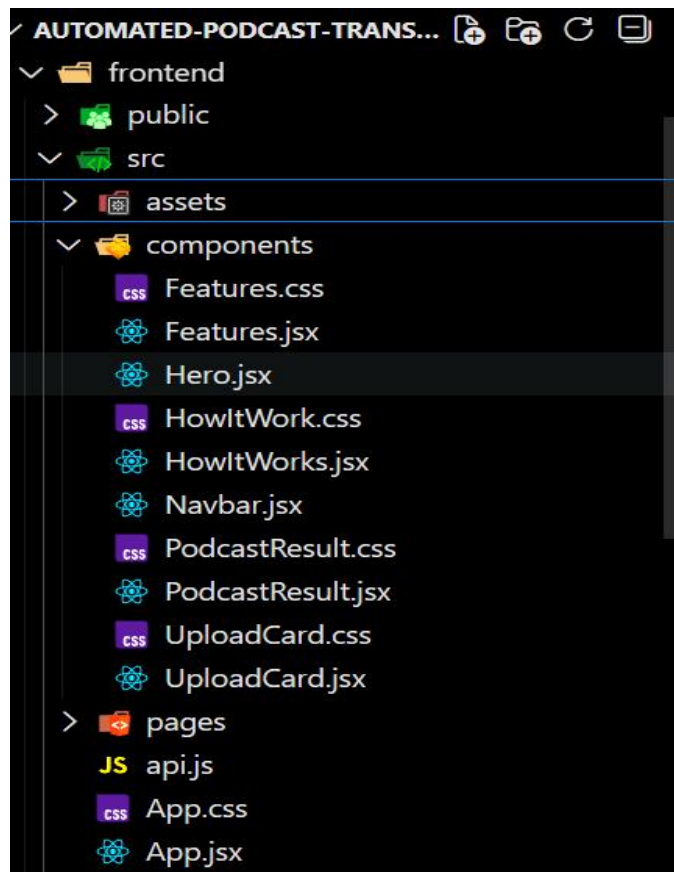
- **app.py – The Orchestrator (Backend API)** This is the central nervous system of the project. It handles the **HTTP requests** from the frontend. When an audio file is uploaded, app.py triggers the sequential pipeline: calling the preprocessing, then transcription, and finally the segmentation modules. It concludes by packaging the final results into a structured **JSON response**.
- **preprocess.py – The Audio Cleaning Unit** Raw audio is often "noisy" or in the wrong format. This module ensures the AI model receives a "clean" signal. It performs:
 - **Format Normalization:** Converting various formats (MP3, AAC, etc.) into a standard .wav.
 - **Noise Reduction:** Filtering background static to improve word accuracy.
 - **Resampling:** Downsampling the audio to exactly **16kHz**, which is the optimal frequency for the Whisper model.
- **transcribe.py – The Speech Recognition Engine** This module interfaces with the **OpenAI Whisper** model. It loads the pre-trained weights and processes the clean audio. Its primary job is to turn acoustic signals into text tokens while preserving **millisecond-accurate timestamps** for every word or phrase detected.

- **segmentation.py – The Semantic Analyzer (Topic Segmentation)** Raw text is hard to read. This module uses NLP to find "semantic boundaries"—points where the speaker



Frontend (React):

- **App.jsx (Root Orchestrator):** The main entry point that manages the global state of the application, including the transition from the "Upload" view to the "Results" view.
- **api.js (Service Layer):** A dedicated module for handling HTTP requests. It uses **Axios** to send the audio binary to the Flask /upload endpoint and receives the processed JSON data..
- **UploadCard.jsx (Input Module):** A specialized component for file ingestion. It handles file validation, drag-and-drop events, and triggers the upload process.
- **PodcastResult.jsx (Data Visualization):** The most complex component. It parses the JSON response to render the segmented timeline, summary cards, and the interactive audio player.



USE CASES

Use Case 1: Search Inside a Podcast (Audio Search Engine)

Description:

This use case enables users to search for any word, phrase, or topic within a podcast and instantly locate where it is mentioned. The system converts the audio into a transcript and indexes the text. When a user enters a search query, the AI retrieves the exact timestamps where the term appears and allows direct navigation to those points in the audio.

Purpose:

To make long-form audio content easily searchable, allowing users to quickly access specific information without listening to the entire episode.

Benefits:

- Helps students instantly locate academic or technical topics
- Assists researchers in retrieving precise information from lengthy discussions
- Enables journalists to find accurate quotes or statements quickly
- Improves user engagement on podcast platforms by offering smart navigation

Use Case 2: Accessibility for Hearing-Impaired Users

Description:

This use case focuses on improving accessibility by converting podcast audio into readable transcripts and concise summaries. Individuals who are deaf or hard-of-hearing can access the same information by reading the text version of the podcast instead of listening to the audio.

Purpose:

To ensure that podcast content is accessible to all users, including those with hearing disabilities or language barriers.

Benefits:

- Provides equal access to content for hearing-impaired users
- Helps non-native English speakers understand the material more easily
- Supports schools and institutions in creating inclusive educational resources
- Expands the potential audience and improves platform accessibility compliance.

ADVANTAGES

- **Automated:** Eliminates the need for manual transcription, saving human effort.
- **Time-efficient:** Processes an hour of audio in a fraction of the time.
- **Scalable:** The pipeline can be adapted to process multiple files simultaneously.
- **User-friendly:** A clean React interface makes complex AI results accessible to non-technical users.

LIMITATIONS

- **High Latency:** Long audio files (1 hour+) require significant time for the Whisper model to process on standard hardware.
- **Hardware Constraints:** Without a high-end GPU, CPU-based inference can be slow.
- **Connectivity:** Currently depends on internet access for certain NLP libraries or API-based modules.
- **Language Accents:** While Whisper is robust, heavy regional accents or noisy backgrounds can still lead to minor word errors.

FUTURE ENHANCEMENTS

- **Real-time Transcription:** Processing audio streams live as the user speaks.
- **Speaker Diarization:** Identifying and labeling "Who spoke when" (e.g., Speaker 1 vs. Speaker 2).
- **Cloud Deployment:** Moving the backend to AWS/Azure for auto-scaling and GPU acceleration.
- **Search within Audio:** Allowing users to type a keyword and jump to that specific moment in the player.

CONCLUSION

- **Problem Solved:** This project addresses the "Information Overload" found in raw audio by converting unstructured speech into a searchable, summarized, and navigable digital asset.
- **Skills Learned:** I gained hands-on experience in **Full-stack AI integration**, signal processing (FFmpeg), Deep Learning (Whisper), and building responsive UIs in React.
- **Industry Relevance:** Automated transcription and summarization are currently multi-billion dollar industries (similar to Otter.ai or Descript). Mastering this pipeline demonstrates a strong grasp of how AI can be applied to solve real-world business problems.